

Question 1

[10]

Define or explain in words the following terms:

- a) Gauss's law
- b) Ampere's law
- c) Magnetic Flux
- d) Potential difference
- e) Dielectric material

Question 2

[20]

- a) If you have a metal sphere with +Q charge enclosed by a pair of hemispherical shells. If you briefly ground the outer shell, what will you observe?
- b) Suppose there are two charges fixed at different points, each exerting a force on a third charge. What will be the force on the third charge?
- c) If you have a charge of 10 nC, how much electric flux will be generated by that charge?
- d) True or false, the electric field vector points in the same direction as the distance vector?
- e) True or false, application of an electric field can lead to the formation of dipoles in a nonpolar material?
- f) What happens to the strength of a magnetic field as you move away from source of current?
- g) What will be the direction of the current that produces a magnetic field in the $-\mathbf{a}_\phi$?
- h) Distinguish between a magnetic force and electric force by listing one difference between the two.
- i) Can a segment on a current carrying conductor exert a force on another segment on the conductor?
- j) Will a pair of parallel lines with current in the same direction experience a force of attraction or repulsion?

Question 3

[10]

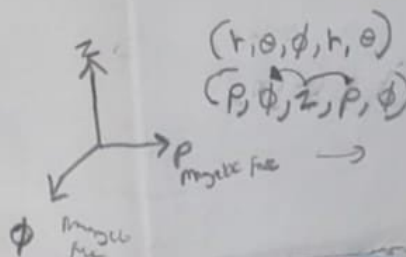
The field for an infinite length line of charge on the z-axis is given by

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho$$

Suppose the charge density $\rho_L = 100 \text{ nC/m}$. Find the work done in moving a 10 nC charge from

$\rho = 3 \text{ m}$ to $\rho = 1 \text{ m}$.

20 kJ



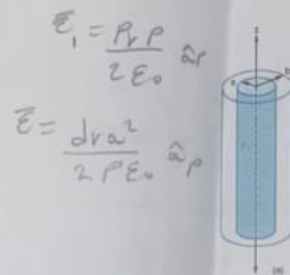


Question 4

Using Biot-Savart law show that a line of current in the $+z$ direction on the z -axis cannot exert a magnetic force on itself. [5]

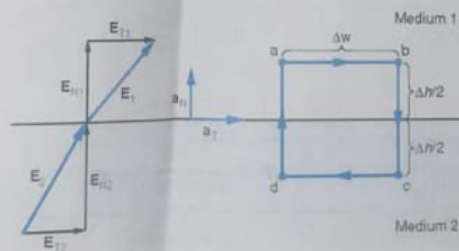
Question 5

Consider an infinitely long coaxial cable as shown in the diagram below. The charge density in the inner cylinder (radius a) is ρ_v and that the outer cylinder (radius b) is grounded. Find the electric flux density in the region $a \leq \rho \leq b$, using Gauss's law. [15]



Question 6

Derive the first boundary condition for dielectric materials oriented as shown below. [20]

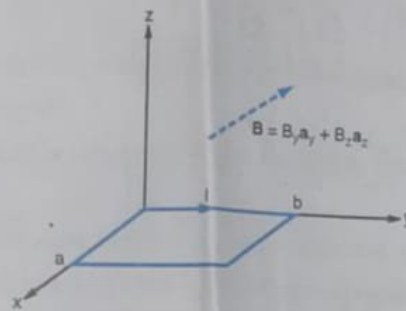


**Question 7**

Derive the magnetic field intensity expression at any point in space resulting from an infinite line of current placed along the z-axis. Using Ampere's law.

Question 9

Calculate the total torque on a current carrying loop in the x-y plane immersed in a uniform magnetic field $\mathbf{B} = B_y \mathbf{a}_y + B_z \mathbf{a}_z$



Useful expressions

$E = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R$	$\mathbf{H} = \int \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$
$E = \frac{\rho_L}{2\pi\epsilon_0 \rho} \mathbf{a}_\rho$	$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$
$E = \frac{\rho_L a h}{2\epsilon_0 (a^2 + h^2)} \mathbf{a}_z$	$\mathbf{H} = \frac{I a^2}{2(h^2 + a^2)^{3/2}} \mathbf{a}_z$
$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$	$I = \int \mathbf{J} \cdot d\mathbf{S}$

